

Legal Agent for Vision Datasets

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Motivation

The Data Privacy Bottleneck

- **The Compliance Gap:** Researchers have models but lack legal expertise for GDPR/CCPA.
- **The Scalability Crisis:** Manual blurring of thousands of hours of video is impossible; static scripts are too "dumb" to handle edge cases.
- **High Stakes:** For institutions like UVA, a single PII leak risks fines, funding loss, and reputational damage.

Background

The Data Paradox: Generative and Embodied AI require vast amounts of real-world data, but strict data privacy remains the "Achilles' heel" of dataset publication.

Limitations of Traditional De-ID: Standard de-identification (De-ID) relies on static scripts (e.g., basic face blurring) that frequently fail on complex, real-world edge cases.

The Agentic Solution: Autonomous AI Agents utilize active reasoning and tool-use, transforming rigid privacy scripts into dynamic decision-making systems capable of reliably handling edge cases.

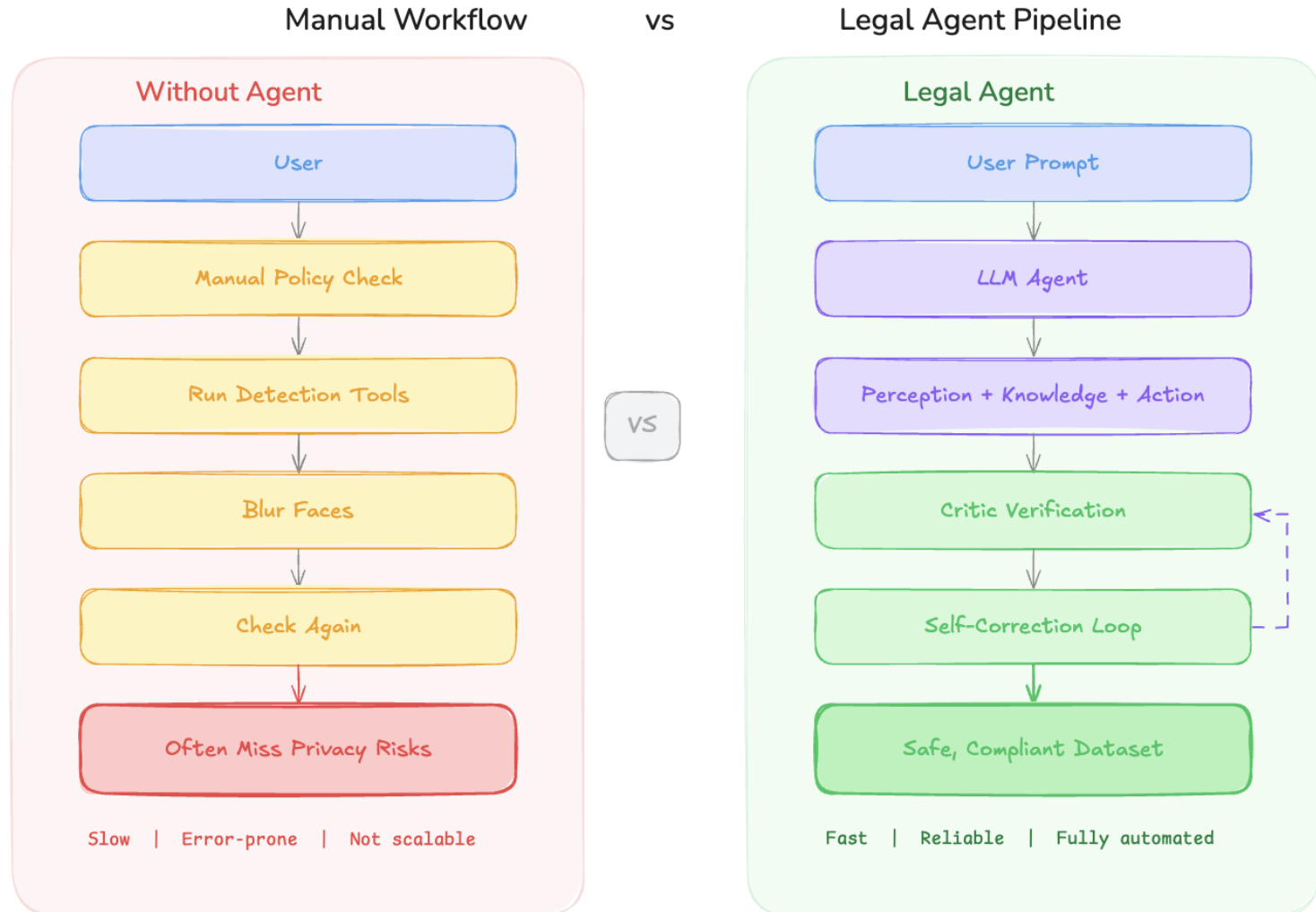
Related Work

- Foundational Vision:** Traditional detectors (YOLO, MTCNN, S3FD) effectively localize PII but lack contextual understanding.
- Privacy vs. Utility:** Research into DP-GANs and k-anonymity highlights the ongoing trade-off between protecting data and maintaining its usefulness.
- Agentic Orchestration:** Frameworks like LangChain and AutoGPT prove LLMs can act as reasoning engines to manage complex, multi-tool workflows.
- The Current Gap:** Existing compliance tools focus heavily on text-based PII, leaving a critical need for your De-ID Agent to handle the complexities of high-dimensional visual data.

Claim / Target Task

- **The Objective:** Solve the "Dataset Release Dilemma" by deploying an autonomous De-ID Agent.
- **The Input:** Raw visual data (image/video) + a defined "Compliance Profile" (e.g., OpenReview standards).
- **The Autonomous Workflow:**
 - **Identify:** Locate all PII, including faces, license plates, and sensitive text.
 - **Anonymize:** Execute a robust, multi-stage obscuration process.
 - **Verify:** Perform adversarial testing to guarantee successful de-identification.
 - **Report:** Output a standardized Privacy Compliance Report for safe dataset release.

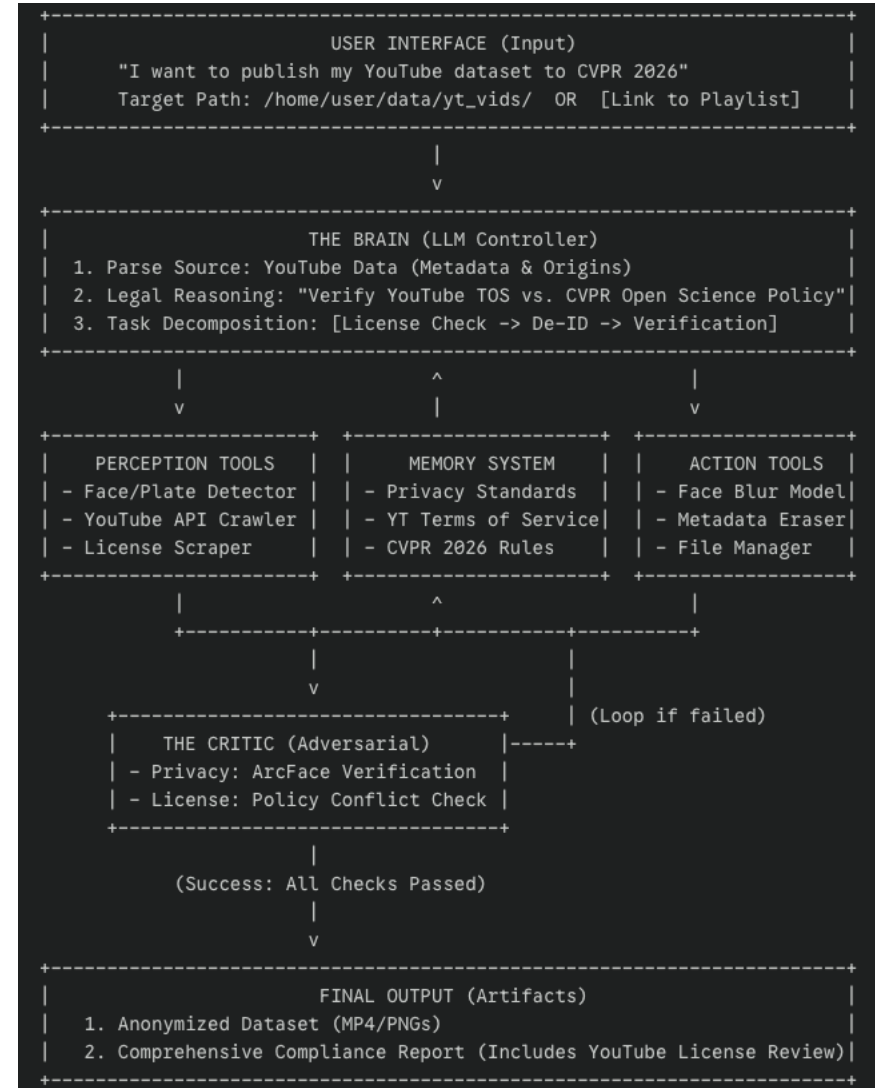
An Intuitive Figure Showing WHY Claim



Proposed Solution

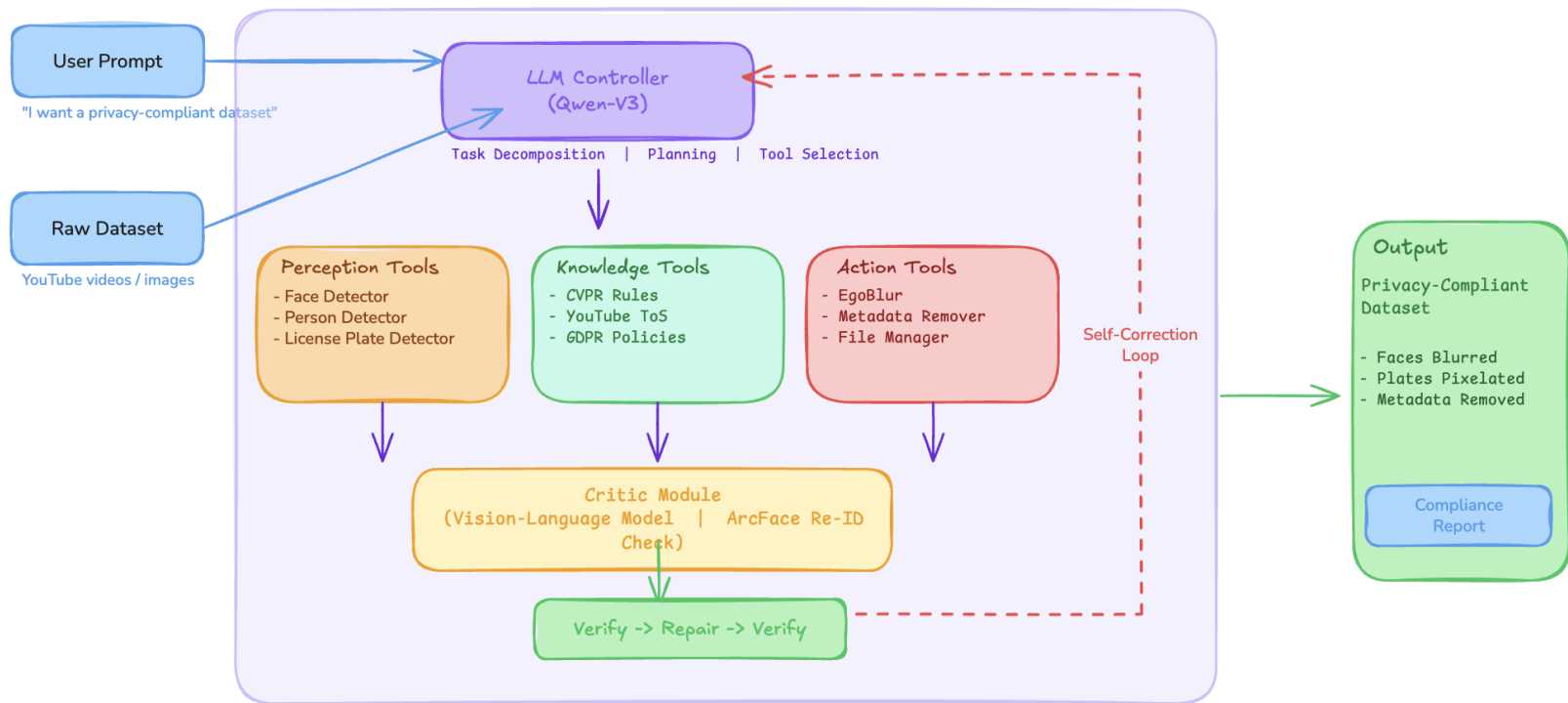
Baseline

- LLM Controller: Qwen-V3 to decompose the uncertain prompt into tasks
- Dataset for demonstration: D-RE15K (videos + images collected from Youtube)
- Collection of Tools
 - Perception: Pretrained Object Detector (faces / persons / plates / sensitive regions)
 - Knowledge: retrieval of platform/conference regulations
 - Action: egoblur + file manager + multiprocessor runner
 - Critic: Vision-language reasoners for verification



Implementation

Legal Agent for Datasets — System Architecture



Data Summary

Dataset : 76 egocentric video clips sourced from YouTube

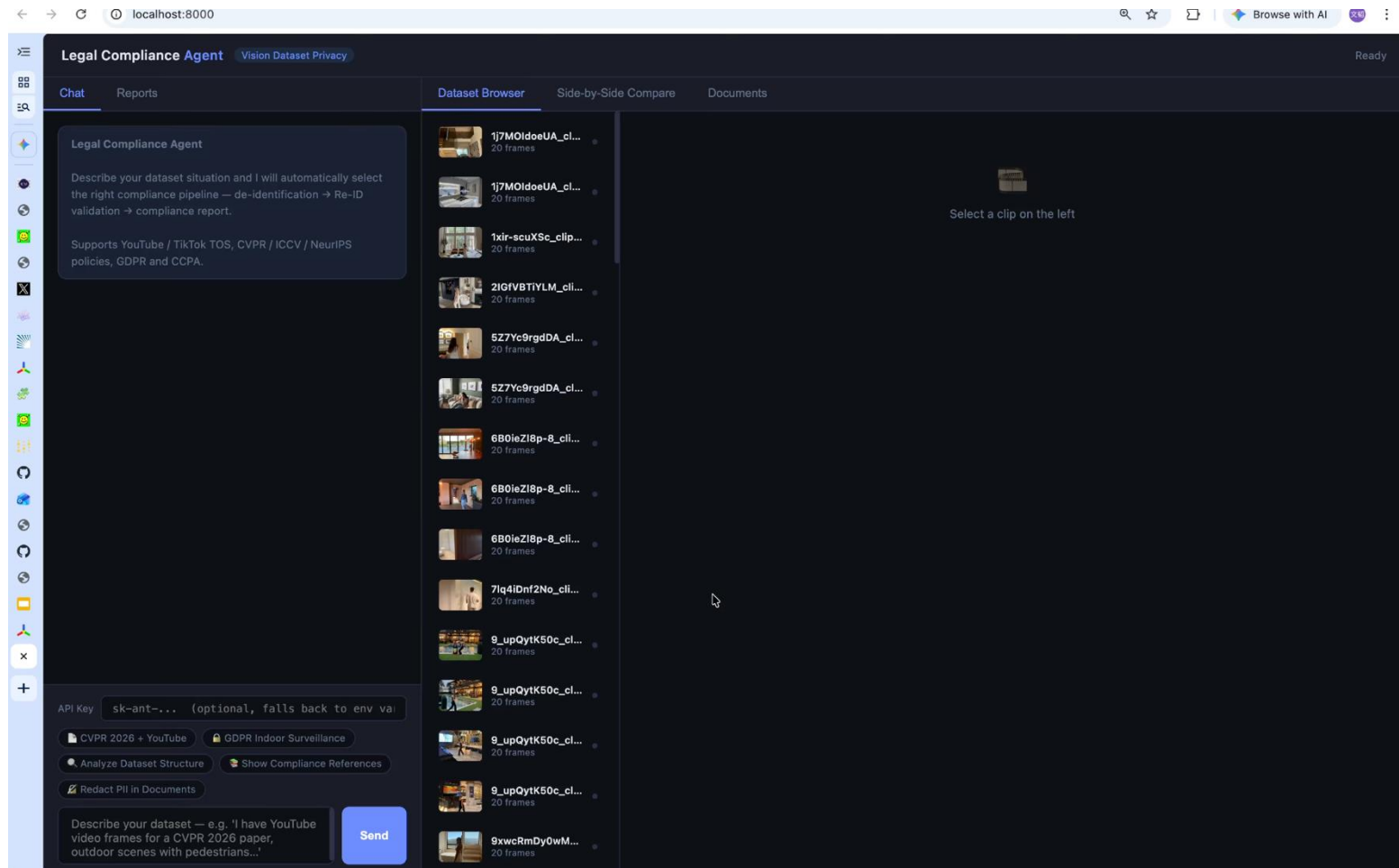
- 1,540 frames total
- Representative of real-world vision dataset collection scenarios

Document Corpus: 5 Markdown documents

- 107 injected PII items across 8 categories
- Categories: Person Name, Email, Phone, IP Address, Physical Address, SSN, Credit Card, Date of Birth

Video Demo

<https://youtu.be/CYngcBjpKdM>



User Cases

Case A: The "Conference Compliance" Request (Uncertain Goal)

•**User Input:** *"I'm finishing a paper for CVPR. I have a folder of raw test videos from YouTube, but I'm worried about the new ethics guidelines. Can you make sure I don't get desk-rejected for privacy issues?"*

•**Agent Reasoning & Action:**

1. **Requirement Extraction:** The Agent calls a `Web_Search_Tool` to retrieve the *CVPR 2026 Author Guidelines* and *YouTube Terms of Service*. It identifies that "De-identification of human subjects" is mandatory.
2. **Uncertainty Resolution:** The Agent scans the dataset and detects that 20% of the videos are "First-Person View" (Ego-centric). It decides to use `egoblur` for these specifically, while using a standard `MTCNN-Face-Blur` for static tripod shots to maintain higher background utility.
3. **Autonomous Tool Call:** It initializes a `Multiprocessor_Runner` to batch process the `D-RE15K` dataset.

•**Agent Result:** The Agent returns a processed dataset and a summary: *"I have anonymized 15,000 frames. Based on CVPR's ethics policy (Section 4.2), I applied heavy Gaussian blur to faces and pixelated license plates. I also stripped EXIF metadata to ensure a double-blind review."*

Case B: The "Recursive Quality Control" (Self-Correction)

•**User Input:** *"Process this indoor surveillance dataset for public release. Keep it as realistic as possible for training, but ensure it's GDPR compliant."*

•**Agent Reasoning & Action:**

1. **Conflict Resolution:** The Agent recognizes a trade-off: GDPR requires "Irreversible Anonymization," but the user wants "Realism."
2. **Action Engine:** It applies a light `Generative_Face_Refacing` tool (replacing real faces with AI-generated ones) rather than simple blurring to preserve facial landmarks for training.
3. **Verification Loop (The Critic):** After processing, the Agent calls a `Face_Recognition_Critic` (ArcFace). The Critic reports: *"Confidence score for Subject_04 is still 0.72 (Identifiable)."*
4. **Self-Correction:** The Agent autonomously re-runs the pipeline for those specific frames with a higher noise seed until the Re-ID confidence drops below 0.1.

•**Agent Result:** A high-fidelity "Synthetic-Face" dataset and a **Validation Log** proving that 100% of the subjects failed the Re-identification test.

Experimental Analysis

- **Re-Identification Rate (Re-ID):** The percentage of faces that can still be matched by ArcFace after anonymization.
- **Structural Similarity Index (SSIM):** Ensuring the "background" of the research data remains high-quality for training despite the foreground masking.

Experimental Results

- All the private information has been removed, both on image and text dataset.
- The SSIM score shows the processed images keep their quality for further research.

Metric	Score
Re-ID Rate	0.00%
Mean SSIM (All)	0.9994
Mean SSIM (Modified)	0.9985

Conclusion and Future Work

- We present a Legal Agent for Vision datasets to lower the barrier for researchers who need to release privacy-compliant datasets.
- Our evaluations on the image dataset and documentations show the effectiveness of our agent.
- In the future, we hope there can be more fine-grained private information detection not limited to research datasets.

References

- [1] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," **IEEE Transactions on Image Processing**, vol. 13, no. 4, pp. 600–612, 2004.
- [2] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in **Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, 2019, pp. 4690–4699.
- [3] European Parliament and Council, "General Data Protection Regulation (GDPR)," Regulation (EU) 2016/679, 2016.
- [4] State of California, "California Consumer Privacy Act (CCPA)," AB-375, 2018.
- [9] CVPR, "CVPR 2026 open science policy: Dataset and benchmark track guidelines," IEEE/CVF, 2026.
- [10] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in **Proc. IEEE International Conference on Image Processing (ICIP)**, 2016, pp. 3464–3468.
- [11] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in **Proc. IEEE International Conference on Image Processing (ICIP)**, 2017, pp. 3645–3649.
- [13] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, "Joint face detection and alignment using multitask cascaded convolutional networks," **IEEE Signal Processing Letters**, vol. 23, no. 10, pp. 1499–1503, 2016.